NYC Air Quality Analysis: Seasonal Trends, District Hotspots, and Predictive Modeling (2009–2023)

BAN 586 Capstone Project — Fall 2025

Vamshi Aitharaju

1. Executive Summary

This project looks at how New York City's air quality has shifted over a fifteen–year period and how different neighborhoods experience pollution differently. I focused on NO_2 and PM2.5 because they had consistent reporting and are two of the most important pollutants for public health. My goal was to understand seasonal behavior, identify districts that repeatedly show higher concentrations, and evaluate whether these patterns could be modeled or forecasted.

From the analysis, NO_2 showed a very clear seasonal pattern, rising sharply every winter due to heating fuel use and higher traffic. PM2.5 barely changed between seasons and behaved more like a steady background pollutant. District–level comparisons showed that parts of Manhattan —especially Midtown—had consistently higher values, while Staten Island tended to remain on the lower end. Hypothesis testing supported these observations, confirming a significant difference between winter and summer NO_2 but not for PM2.5. Forecasting results from both Excel and Python pointed toward gradual improvement in both pollutants through 2027.

To compare predictive techniques, I used RapidMiner to build a Decision Tree, Linear Regression model, and Random Forest model. The Random Forest model performed the most consistently, while the Decision Tree was the easiest to interpret. These findings support the idea that NYC should continue winter–focused mitigation strategies and expand monitoring in neighborhoods that repeatedly show higher concentrations.

2. Project Statement

The purpose of this project is to explore how NYC's air pollution changes by season and location, and whether those patterns can help with planning and forecasting. The analysis focuses on three main questions:

1. Which community districts show the highest levels of NO_2 and PM2.5?

2. How do seasonal changes impact pollution patterns across districts?

3. Can forecasting methods provide a reasonable sense of how pollution values may shift in the next few years?

To answer these questions, I performed data cleaning, exploratory analysis, statistical testing, forecasting, and predictive modeling.

3. Data and Methods

The dataset used in this project was published by the NYC Department of Health and Mental Hygiene and includes measurements collected between 2009 and 2023 across 59 community districts.

Data Cleaning:

- Dropped extra metadata fields that were not needed for analysis.

- Removed early years that did not align with the main date ranges.

- Extracted community district identifiers from text fields.

- Reorganized data into Winter, Summer, and Annual Average subsets.

- Standardized units and rounded numerical values.

Methods:

I used Excel for cleaning and general descriptive work, Python for hypothesis testing and forecasting, RapidMiner for machine–learning models, and Power BI to explore visual trends. This combination allowed me to verify patterns from multiple perspectives.

4. Models and Analysis

Exploratory Analysis:

NO_2 rises strongly in winter and drops in summer. PM2.5 remains steady. Midtown consistently reports higher values; Staten Island reports lower ones. Correlation between pollutants was close to zero.

Hypothesis Testing:

- Winter vs. Summer NO_2: significant difference.

- Winter vs. Summer PM2.5: not significant.

Forecasting:

Excel and Python forecasts pointed toward a continued downward trend through 2027.

RapidMiner Models:

- Decision Tree ($R^2 \approx 0.78$): intuitive structure; major splits based on pollutant and season.

- Random Forest ($R^2 \approx 0.78$): stable across districts.

- Linear Regression ($R^2 \approx 0.69$): helpful baseline model.

5. Validation and Testing

A 70/30 training–testing split was used to validate all models. R^2 values were compared, and forecast trends were checked across Excel and Python. Predictions aligned with expected seasonal behavior, particularly NO_2 winter spikes.

Random Forest performed most reliably, while Decision Tree was most interpretable. Linear Regression provided a helpful comparison point.

6. Results and Recommendations

Key Results:

- NO_2 consistently increases during winter.

- PM2.5 stays steady across seasons.

- Manhattan districts form persistent hotspots.

- Staten Island maintains lower pollution levels.

- Long–term trends show gradual improvement.

- NO_2 and PM2.5 act independently.

Recommendations for NYC:

- Strengthen winter NO_2 reduction measures.

- Increase air–quality sensors in high–risk districts.

- Use interactive dashboards and forecasts for planning.

Recommendations for Researchers:

- Include weather and traffic variables.

- Use more granular timeframes.

- Experiment with ARIMA, LSTM, or advanced prediction models.

7. Conclusion

This project demonstrates that NYC's air quality follows clear seasonal and district–level patterns backed by statistical tests, forecasting, and machine–learning results. NO_2 rises mainly due to winter heating and traffic, while PM2.5 stays relatively stable across seasons. Although both pollutants show improving trends, targeted policies are still needed in neighborhoods with persistently higher concentrations.

8. Acknowledgements

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9. References

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